MIL-B-52118D
31 January 1975
SUPERSEDING
MIL-B-52118C
8 December 1969

MILITARY SPECIFICATION

BALK, DECK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers deck balk which is a component of military bridges.

1.2 Classification. The balk shall be of the following types, as specified (see 6.2):

Type I - Short
Type II - Tapered
Type III - Normal

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-16173	- Corrosion Preventive Compound, Solven Cutback, Cold-Application.

FSC 5420

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

TA13208E9880	- Balk, Deck, Normal.
TA13208E9881	- Balk, Deck, Short.
TA13208E9886	- Balk, Deck, Tapered.
D13217E7892	- Balk, Deck, Short: Packing of.
D13217E7893	- Balk, Deck, Normal: Packing of.
D13217E7894	- Balk, Deck, Tapered: Packing of.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.1	- Mild Steel Covered Arc-Welding Electrodes.
A5.10	- Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
A5.18	- Mild Steel Electrodes for Metal-Arc Welding.
D1.1	- Structural Welding Code.

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(Application for copies should be addressed to the American Welding Society, Inc., 2501 N. W. Seventh Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The balks shall be as shown on TA13208E9880, TA13208E9881, and TA13208E9886 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the balk or components. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish two balks or components, as applicable, for inspection within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods will produce balks or components that comply with the requirements of this specification. One balk shall be complete with painting, stenciling, and die stamping and the other balk shall be complete but shall only have the necessary prime coat of paint applied. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Welding electrodes.

3.4.1 For steel. Welding electrodes for steel shall conform to AWS A5.1 or A5.18, Classification E60 or E70 series, size as required.

3.4.2 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, Classification ER4043, size as required.

3.5 Identification marking. The balk or component, as applicable, shall be metal die stamped as shown on the applicable drawing, and markings shall be legible after all painting has been completed.

3.6 Cleaning, treatment, and painting. Cleaning, treatment, and painting shall be as follows:

(a) The backup plates, lugs, handles, and the balk extrusions (where the backup plates, lugs, and handles come in contact with the balk extrusions) shall be cleaned and treated in accordance with MIL-T-704, Type B.
(b) The shoe assembly of the tapered balk shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B, prior to assembling in the top and bottom balk extrusions.
(c) The exterior of each completely assembled balk shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B.
(d) All components, when procured separately from balk, except threads on plugs and threads on bolts of end lug assemblies, shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B.

3.7 Stenciling. The nomenclature, National Stock Number, weight, and cube shall be stenciled on each deck balk. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Type I, balk, deck, short. The Type I balk shall be as shown on TA13208E9881 and as specified herein. The welds on the balk shall not leak when subjected to an internal air pressure of 3 pounds per square inch (psi).

3.9 Type II, balk, deck, tapered. The Type II balk shall be as shown on TA13208E9886 and as specified herein. No tests are required on this balk.

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3.10 Type III, balk, deck, normal. The Type III balk shall be as shown on TA13208E9880 and as specified herein. When supported on the horizontal leg of the end lugs and a downward vertical load of 40,000 pounds is applied to one of the interior lugs, the balk shall not fail or show any visible damage such as cracks or permanent deformation in the welds, extruded sections, lugs, bolts, or base metal adjacent to the welds. The welds on the balk shall not leak when subjected to an internal air pressure of 3 psi.

3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8):

ITEM	DESCRIPTION	IDENTIFICATION	QTY
1	Inspection Equipment for Balk, Deck	Quality Assurance Pamphlet, P715-107	1

3.12 Workmanship. The fabricated balk shall be free of sharp edges, sliver or burrs.

3.12.1 Steel fabrication. Corners shall be square and true. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks.

3.12.2 Aluminum fabrication. Before any work or layout is done, aluminum shall be straightened. Straightening shall be done by methods that will not result in injury to the metal. Care shall be used in handling aluminum during fabrication to avoid scratching or damage to edges. Finished members shall be free from laminations, kinks, bends, scratches or winds. All bends shall be made to a radius which will not cause cracking or undue stress in the extreme outer fibers. Except for those of a minor nature, bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Sharp instruments shall not be used for making bend lines. Aluminum 3/8 inch thick and over shall be cut by routing or sawing. Aluminum under 3/8 inch thick may be cut by shearing.

3.12.3 Riveting. Holes for countersunk heads shall be of such a depth and angle that the head shall fill the hole without projecting or recessing. Heads of driven rivets shall be uniform in shape and concentric with the rivet shank axis, and shall have full bearing on the surfaces of the joined members. Rivets shall fill the holes in the joined members. Caulking of rivet heads shall not be permitted. Components to be riveted shall fit correctly prior to riveting to assure that rivets are not placed in tension. Rivets that are loose or have a head deficient in size or are misshapen in any way shall be considered defective and shall be removed without injury to the joined metal.

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Where it is practical, rivets shall be driven by direct-action rivet machines. In lieu of direct-action rivet machines, pneumatic rivet hammers may be used. Excessive air driving pressures or overdriving of rivets resulting in damage to the joined members shall not be permitted.

3.12.4 Welders and welding.

3.12.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.
ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

3.12.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process. Steel welding process shall conform to AWS D1.1 - Structural Welding Code. Aluminum welding process shall conform to ASME Boiler and Pressure Vessel Code, Section IX Part B. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practical. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire-brushed clean.

3.12.4.2.1 Steel welding. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed.

3.12.4.2.2 Aluminum welding. Aluminum members to be welded may be preheated for welding to a temperature of not more than 400° F. Top and bottom extrusions shall be butt-welded by the automatic-machine method. The weld shall be made at a uniform rate in one pass and shall be backed up for its entire length. Skips and blow-through holes in automatic-machine-made welds may be corrected by manual welding if the sum of the length of skips and blow-through holes is not more than 2 percent of the total length of the weld, and if the length of any one skip or blow-through hole is not more than 1 inch. The ends of the skips and the total length of the blowholes shall be chipped to parent metal before welding. Aluminum welds may be ground, filed, wire-brushed, or chipped but shall not be hammered.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provision (SQAP) shown on the drawings. The drawings specify the characteristics requiring inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6).

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction deck balks or components shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects in either deck balk or component shall be cause for rejection of both balks.

4.3.2 Tests. The preproduction deck balks shall be tested as specified in 4.5.2 as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. The lot size shall not exceed 50 balks.

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4.4.2 Examination.

4.4.2.1 Individual. Each balk shall be examined for conformance to the dimensions covered by the SQAP shown on the drawings under "Major-100 percent inspection" (see 6.6.1). Unless otherwise specified (see 6.2), Government-loaned inspection equipment (see 3.11) shall be used in accordance with Government QA Pamphlet included in the TA package (see 3.1) to inspect these dimensions. Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Samples of normal balks selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. Samples of short balk selected in accordance with 4.4.1 shall be as specified in 4.5.2.2. AQL shall be 2.5 percent defective.

4.5.1 Examination. The balk or component, as applicable, shall be examined as specified herein for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Workmanship not as specified.
104. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
105. Radii on interior and end lugs not as specified.
106. Riveting not as specified.
107. Chamfers for O-rings not as specified.
108. Installation of O-rings not as specified.

Minor

201. All dimensions not designated major on the SQAP, not as specified.
202. Identification marking, incorrect, missing, or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

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4.5.2 Tests.

4.5.2.1 Load. Support the normal deck balk on the horizontal leg of the end lugs and apply a downward vertical load of 40,000 pounds to one of the interior lugs for a minimum of 5 minutes. Any cracks or permanent deformation in the welds, extruded sections, lugs, bolts, or base metal adjacent to the welds shall constitute failure of this test.

4.5.2.2 Leakage. Subject the normal and short deck balk to an internal air pressure of 3 psi applied through the end-plug hole and submerge the balk in water. Any air leak through any weld shall constitute failure of this test. This test shall be performed prior to final painting and after load testing of the balk.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109.	Threads of separately procured components not coated with preservative as specified.
110.	Balk not bundled as shown on drawing.
111.	Balk components of unlike description packed together.
112.	Containers not as specified for Level A or B.
113.	Blocking and bracing of components within the container not adequate to prevent free movement for Level A or B.
114.	Strapping on containers not zinc coated for Level A.
115.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to

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starting production packaging, that the applied preservation, packing, and marking comply with the preparation for delivery requirements of this specification. Examination shall be that specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction balk or the production balk. If the preproduction model balk is utilized, any preservation and packing shall be removed by the supplier at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model balk and the production balk.

5.2 Preservation (see 6.7).

5.2.1 Balk. The balk, cleaned, treated, and painted as specified shall not require additional preservation.

5.2.2 Components. When balk components are procured separately from balk, the threads on plugs and threads on bolts of end lug assemblies shall be coated with preservative conforming to MIL-C-16173, Grade 1.

5.3 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Balk. The balk shall be prepared for shipment in bundles as shown on Drawings D13217E7892, D13217E7893, or D13217E7894, as applicable.

5.3.1.2 Components. Balk components of like description, procured separately from the balk, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. The components shall be blocked and braced within the box to prevent free movement. Box closure and strapping shall be in accordance with the appendix to the box specification. Unless otherwise specified (see 6.2) strapping shall conform to QQ-S-781, Finish B.

5.3.2 Level B.

5.3.2.1 Balk. The balk shall be prepared for shipment as specified in 5.3.1.1.

5.3.2.2 Components. Balk components of like description, procured separately from the balk, shall be packed together in a close-fitting box conforming to PPP-B-601, Domestic Type, style optional, or PPP-B-636, V3c, style optional. The components shall be blocked and braced within the box to prevent free movement. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

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5.3.3 Level C.

5.3.3.1 Balk. The balk shall be prepared for shipment as specified in 5.3.1.1.

5.3.3.2 Components. Balk components of like description, procured separately from the balk, shall be packed together in a manner to assure carrier acceptance and safe delivery to destination at lowest rating in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Each bundle of balk and each box of components shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The balk is used as a deck component of the superstructure for floating and short fixed-span bridges.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of balk required (see 1.2).
(c) Time frame required for submission of the preproduction model (see 3.2.1).
(d) When Government-loaned property is not required (see 3.11 and 4.4.2.1).
(e) Time frame required for submission of the preproduction pack (see 5.1).
(f) Level of packing required (see 5.3).
(g) When strapping other than as specified is required (see 5.3.1.2 and 6.9).

6.3 Preproduction model. Any changes or deviations of production deck balk or component from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish deck balk conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, pack, and mark the balk or components in accordance with this specification.

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6.5 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum,lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.6 Definitions.

6.6.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Preservation. Levels of preservation have not been included as the preservation specified is the minumum acceptable for any level.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.11, when applicable.

6.9 Strapping. Army policy requires the use of organic coated (Finish A) strapping for Army-sponsored shipments.

Custodian:

Army - ME

Preparing activity:

Army - ME

Review activity:

Army - SM

User activity:

Navy - MC

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-52118D Balk, Deck

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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